

- 5 (a) (i) ratio of charge (on body) and its potential
(do not allow reference to plates of a capacitor) B1 [1]
- (ii) (potential at surface of sphere $\Rightarrow V = Q / 4\pi\epsilon_0 r$ M1
 $C = Q / V = 4\pi\epsilon_0 r$ A0 [1]
- (b) (i) $C = 4 \times \pi \times 8.85 \times 10^{-12} \times 0.36$
 $= 4.0 \times 10^{-11} \text{ F (allow 1 s.f.)}$ A1 [1]
- (ii) $Q = CV$
 $= 4.0 \times 10^{-11} \times 7.0 \times 10^5$
 $= 2.8 \times 10^{-5} \text{ C}$ A1 [1]
- (c) plastic is an insulator / not a conductor / has no free electrons B1
charges do not move (on an insulator) B1
either so no single value for the potential
or charge cannot be considered to be at centre B1 [3]
- (d) *either* energy $= \frac{1}{2}CV^2$ *or* energy $= \frac{1}{2}QV$ and $C = Q/V$ C1
energy $= \frac{1}{2} \times 4 \times 10^{-11} \times \{(7.0 \times 10^5)^2 - (2.5 \times 10^5)^2\}$ C1
 $= 8.6 \text{ J}$ A1 [3]

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6 (a) unit of magnetic flux density / magnetic field strength

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